# C10 Using Resources – Knowledge Organiser: Self-Test

*AQA GCSE Chemistry • Set 1: Sustainability and Water* **Paper 2**

| # | Question | Attempt 1 | Attempt 2 | Attempt 3 | Answer |
|---|---|---|---|---|---|
| 1 | Sustainable development | | | | Meets present needs without harming future generations |
| 2 | Finite resource | | | | Will run out |
| 3 | Renewable resource | | | | Can be replaced |
| 4 | Potable water | | | | Water safe to drink |
| 5 | Why potable water is not pure | | | | Contains dissolved substances |
| 6 | Two stages to treat fresh water | | | | Filtration, sterilisation |
| 7 | Sterilising agents | | | | Chlorine, ozone, UV light |
| 8 | Methods of desalination | | | | Distillation, reverse osmosis |
| 9 | Why desalination is expensive | | | | Needs large amounts of energy |
| 10 | Waste water treatment stages | | | | Screening, sedimentation, anaerobic digestion of sludge, aerobic treatment of effluent |

*Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.*

# C10 Using Resources – Knowledge Organiser: Self-Test

*AQA GCSE Chemistry • Set 2: Metal Extraction and Life Cycle Assessment* **Paper 2**

| # | Question | Attempt 1 | Attempt 2 | Attempt 3 | Answer |
|---|---|---|---|---|---|
| 1 | Phytomining | | | | Plants absorb metal compounds; burned to ash |
| 2 | Bioleaching | | | | Bacteria produce leachate containing metal compounds |
| 3 | Why new extraction methods are needed | | | | High-grade ores are running out |
| 4 | How copper is got from its compounds | | | | Displacement with scrap iron, or electrolysis |
| 5 | Life cycle assessment stages | | | | Raw materials, manufacture, use, disposal |
| 6 | Why LCAs can be biased | | | | Some values are judgements |
| 7 | Benefit of reducing, reusing and recycling | | | | Saves resources, energy and waste |
| 8 | How glass bottles are recycled | | | | Crushed and melted |
| 9 | How metals are recycled | | | | Melted and recast |
| 10 | Why recycling metals saves energy | | | | Less mining and extraction |

*Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.*

# C10 Using Resources – Knowledge Organiser: Self-Test

*AQA GCSE Chemistry • Set 3: Corrosion and Alloys* Paper 2 Triple only

| # | Question | Attempt 1 | Attempt 2 | Attempt 3 | Answer |
|---|---|---|---|---|---|
| 1 | Corrosion | | | | Destruction of a material by reaction with its environment |
| 2 | What rusting needs | | | | Air (oxygen) and water |
| 3 | Barrier methods | | | | Paint, grease, plastic coating, electroplating |
| 4 | Sacrificial protection | | | | More reactive metal corrodes instead |
| 5 | Galvanising | | | | Coating with zinc |
| 6 | Bronze | | | | Copper + tin |
| 7 | Brass | | | | Copper + zinc |
| 8 | Purity of 24 carat gold | | | | 100% |
| 9 | Steel | | | | Iron + carbon (and other metals) |
| 10 | Stainless steel | | | | Contains chromium and nickel; resists corrosion |

*Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.*

# C10 Using Resources – Knowledge Organiser: Self-Test

*AQA GCSE Chemistry • Set 4: Ceramics, Polymers and Composites* Paper 2 Triple only

| # | Question | Attempt 1 | Attempt 2 | Attempt 3 | Answer |
|---|---|---|---|---|---|
| 1 | Soda-lime glass is made from | | | | Sand, sodium carbonate, limestone |
| 2 | Borosilicate glass is made from | | | | Sand + boron trioxide |
| 3 | How clay ceramics are made | | | | Shape wet clay, heat in a furnace |
| 4 | Why LD and HD poly(ethene) differ | | | | Different catalysts and conditions |
| 5 | Thermosoftening polymers | | | | Soften when heated (no cross-links) |
| 6 | Thermosetting polymers | | | | Do not melt (cross-links) |
| 7 | Composite | | | | Reinforcement held in a matrix (binder) |
| 8 | Examples of composites | | | | Fibreglass, carbon fibre, concrete |
| 9 | High carbon steel | | | | Strong but brittle |
| 10 | Use of aluminium alloys | | | | Aircraft (low density) |

*Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.*

# C10 Using Resources – Knowledge Organiser: Self-Test

*AQA GCSE Chemistry • Set 5: The Haber Process and Fertilisers* **Paper 2** **Triple only**

| # | Question | Attempt 1 | Attempt 2 | Attempt 3 | Answer |
|---|---|---|---|---|---|
| 1 | Haber process makes | | | | Ammonia |
| 2 | Raw materials | | | | Nitrogen (air), hydrogen (natural gas) |
| 3 | Equation | | | | $N_2 + 3H_2 \rightleftharpoons 2NH_3$ |
| 4 | Conditions | | | | 450 °C, 200 atm, iron catalyst |
| 5 | How ammonia is removed | | | | Cooled so it liquefies |
| 6 | What happens to unreacted gases | | | | Recycled |
| 7 | Elements in NPK fertilisers | | | | Nitrogen, phosphorus, potassium |
| 8 | Source of potassium | | | | Potassium chloride and sulfate (mined) |
| 9 | Phosphate rock is treated with | | | | Nitric, sulfuric or phosphoric acid |
| 10 | Ammonia + nitric acid | | | | Ammonium nitrate |

*Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.*